

Background on causality

Background on causality

Consider the following graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$\mathcal{V}$: vertices, $\mathcal{E}$: edges

Path: $D \leftarrow B \rightarrow E \leftarrow C$

Directed path:

$A \rightarrow B \rightarrow E \rightarrow F \rightarrow G$

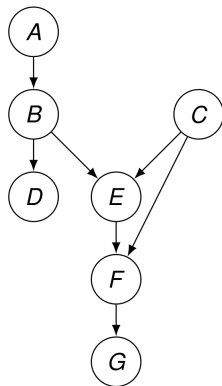
Parents: $Pa(E) = \{B, C\}$

Ancestors: $An(E) = \{A, B, C, E\}$

Children: $Ch(C) = \{E, F\}$

Descendants:

$De(C) = \{C, E, F, G\}$



Background on causality

Compatibility

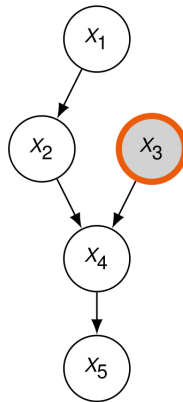
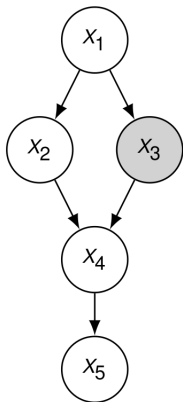
We say that a distribution $P(\mathcal{V})$ is compatible with (or Markov relative to) a DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$$P(\mathcal{V}) = \prod_{X \in \mathcal{V}} P(X | Pa(X))$$

Factorization of $P(A, B, C, D, E, F, G)$ w.r.t. $\mathcal{G}$

$$\underbrace{P(A)}_{\text{no parents}} \cdot \underbrace{P(B | A)} \cdot \underbrace{P(C)}_{\text{no parents}} \cdot \underbrace{P(D | B)} \cdot \underbrace{P(E | B, C)} \cdot \underbrace{P(F | C, E)} \cdot \underbrace{P(G | F)}$$

Background on causality



Background on causality

Truncated factorization

If we intervene on a subset $S \subset \mathcal{V}$, then

$$P(V_1 = v_1, \dots, V_d = v_d | do(S = s)) = \prod_{i \notin S} P(V_i = v_i | V_{pa_i})$$

if $v_1, \dots, v_d$ are values consistent with the intervention

Causal inference

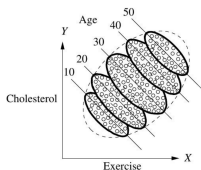
Definition

The causal effect $P(y|do(x))$ from a causal graph $\mathcal{G}$ is identifiable if $P(y|do(x))$ can be computed uniquely from observational data.

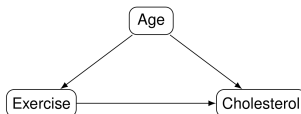
Aim of causal inference : identify effects

Causal inference

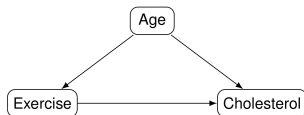
In a study, we measure weekly exercise and cholesterol levels for various age groups



What is the effect of exercise on cholesterol ?



Causal inference



$$\begin{aligned}P(a, e, c) &= P(a)P(e|a)P(c|a, e) \\P(a, c|do(e)) &= P(a)Pr(c|a, e) \\P(c|do(e)) &= \sum_a P(a)P(c|a, e)\end{aligned}$$

Causal inference

Theorem (identifiability in Markovian models)

Given a causal graph $\mathcal{G}$. We assume that the distribution $P(\mathcal{V})$ is compatible with this graph.

The causal effect $P(y|do(x))$ is identifiable whenever $\{X \cup Y \cup pa(X)\} \subset \mathcal{V}$, and is given by the direct causes adjustment:

$$P(y|do(x)) = \sum_{z \in pa(X)} P(y|x, z)P(z)$$

or if the parents set is empty

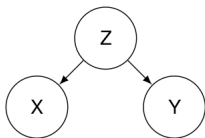
$$P(y|do(x)) = P(y|x)$$

Causal inference

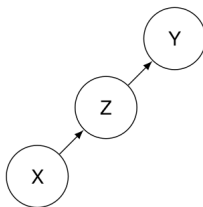
Beyond naive adjustment

- ▶ Is it possible to find a smaller adjustment set?
- ▶ Several criteria used to decide which choice of variables to adjust for to identify a causal effect.
- ▶ A popular one is backdoor based on the concept of d-separation

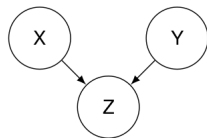
Causal inference



Fork



Chain



v-structure

The concept of d separation

Blocked paths

A path is said to be blocked by a set of vertices Z if:

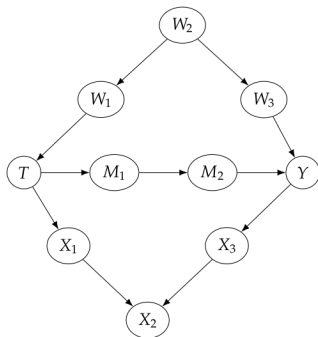
- ▶ it contains a chain $A \rightarrow B \rightarrow C$ or a fork $A \leftarrow B \rightarrow C$ and $B \in Z$, or
- ▶ it contains a collider $A \rightarrow B \leftarrow C$ such that no descendant of B is in Z

The concept of d separation

Definition

Two (sets of) nodes X and Y are d -separated by a set of nodes Z if all of the paths between (any node in) X and (any node in) Y are blocked by Z . We denote $X \perp\!\!\!\perp_G Y | Z$

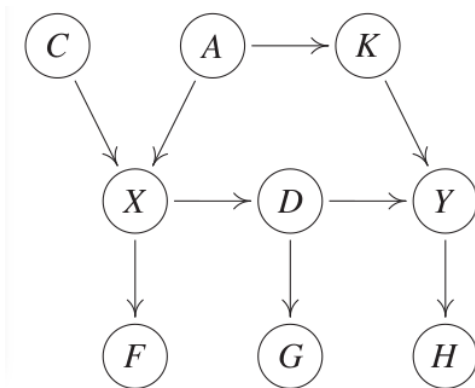
The concept of d separation



Are T and Y d separated by

- ▶ the empty set?
- ▶ $\{W_2\}$?
- ▶ $\{W_2, M_1\}$?
- ▶ $\{W_1, M_2\}$?
- ▶ $\{W_1, M_2, X_2\}$?

The concept of d separation



For this DAG : $C \perp\!\!\!\perp_G G | \{X\}$ and $C \not\perp\!\!\!\perp_G G | \{X, H\}$

Causal inference

The backdoor criterion

- ▶ A set Z satisfies the back-door criterion if:
 - ▶ Z contains no descendant of X , and
 - ▶ Z blocks every back-door path from X to Y , meaning every path that starts with an arrow into X ($X \leftarrow \dots$).
- ▶ If so, then

$$P(y|do(x)) = \sum_z P(y|x, z)P(z).$$

Causal inference

Beyond backdoor

- ▶ In whole generality use the three rules of Pearl's do calculus
- ▶ The ID algorithm

A case Study: Hotel Booking Cancellations

Example extracted from the DoWhy library

- ▶ Scientific question : estimate the impact of assigning a different room as compared to what the customer had reserved on Booking Cancellations.
- ▶ Gold standard : use experiments such as Randomized Controlled Trials wherein each customer is randomly assigned to one of the two categories i.e. each customer is either assigned a different room or the same room as he had booked before
- ▶ What if we cannot intervene or its too costly too perform such an experiment?
- ▶ Can we somehow answer our query using only observational data or data that has been collected in the past?

We shall answer these questions using causality and the dowhy library!

A case Study: Hotel Booking Cancellations

Description of the Dataset

- ▶ Booking information for a city hotel and a resort hotel taken from a real hotel in Portugal
- ▶ Includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things
- ▶ All personally identifying information has been removed from the data

Reference of the study case¹

¹<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

Challenges and principles

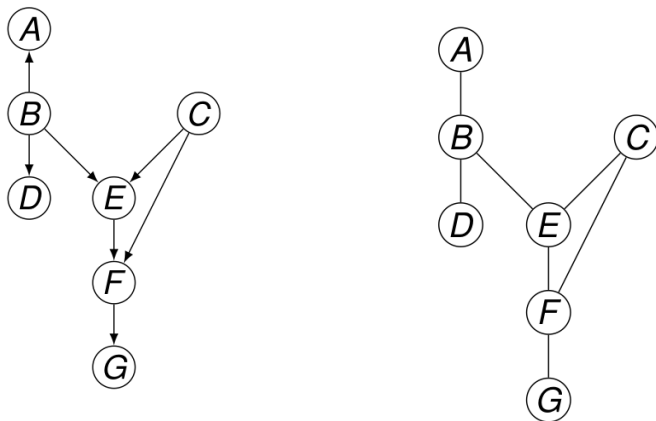
- ▶ In general, causal discovery from observational data is not possible.
- ▶ But it is possible under additional assumptions.
- ▶ Several approaches in the literature
 - ▶ **Constraint based methods** : run local tests of independence to create constraints on space of possible graphs.
 - ▶ **Noise based methods** : find footprints in the noise that imply causal asymmetry.
 - ▶ ...

Principles of constrain based methods

Main steps

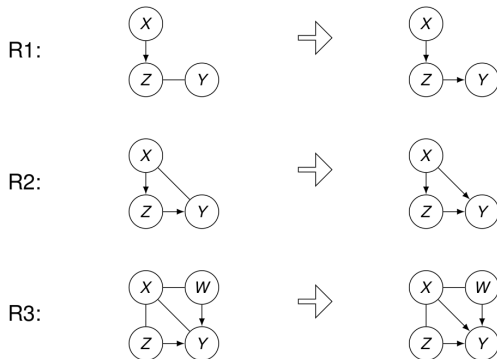
- ▶ Find **skeleton**
- ▶ Find **v-structures**
- ▶ Orient other edges using **basic rules**

Principles of constraint-based methods



a DAG and its corresponding **skeleton**

Principles of constraint-based methods



Basic rules

Principles of constraint-based methods

Algorithm 1 SGS

Input: $P(\mathcal{V})$

Output: CPDAG $\mathcal{G}^*$

- 1: Form the complete undirected graph $\mathcal{G}^*$ on vertex set $\mathcal{V}$
- 2: **for** all $X - Y$ in $\mathcal{G}^*$
and subsets $S \subseteq \mathcal{V} \setminus \{X, Y\}$ **do**
- 3: **if** $\exists S \subseteq \mathcal{V} \setminus \{X, Y\}$ such that $X \perp\!\!\!\perp_P Y \mid S$ **then**
- 4: Delete edge $X - Y$ from $\mathcal{G}^*$
- 5: **end if**
- 6: **end for**
- 7: **for** all $X - Z - Y$ in $\mathcal{G}^*$ such that $X \notin \text{Adj}(Y, \mathcal{G})$ **do**
- 8: **if** $\nexists S \subseteq \mathcal{V} \setminus \{X, Y\}$ such that $Z \in S$ and $X \perp\!\!\!\perp_P Y \mid S$ **then**
- 9: Orient $X \rightarrow Z \leftarrow Y$ in $\mathcal{G}^*$
- 10: **end if**
- 11: **end for**
- 12: Recursively apply rules R1-R3 until no more edges can be oriented
- 13: **Return** $\mathcal{G}^*$

The PC algorithm

- ▶ PC algorithm : optimized version of SGS
- ▶ Infer causal structure with the PC algorithm?
 - ▶ Infer mutual dependencies between variables : **skeleton of the causal graph**
 - ▶ Distinguish between causes and effects : **orientation of the v-structures of the causal graph**

Cause or consequence?

- ▶ Can we distinguish cause from effect?
- ▶ That is distinguish between these two causal graphs

$$X \rightarrow Y$$

or

$$Y \rightarrow X$$

using observational data.

Not always possible!

Cause or consequence?

The example of linear structural equation [f linear]

X cause Y if there exists $a \in \mathbb{R}$, ε^Y s.t.

$$Y = aX + \varepsilon^Y, X \perp \varepsilon^Y.$$

Distinguish cause from consequence? [Shimizu et al., 2006]

Assume that $Y = aX + \varepsilon^Y, X \perp \varepsilon^Y$ where all r.v. are continuous.
Then

$$\exists b \in \mathbb{R}, \varepsilon^X \text{ s.t. } X = bY + \varepsilon^X, Y \perp \varepsilon^X$$

iff (X, ε^X) are Gaussian random variables.

Existence of a non-linear extension of this result.

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Noise based algorithm

Theorem (LiNGAM)

Assume a linear SCM with graph $G = (V, E)$ and a compatible distribution $P(V)$ such that for all $Y \in V$

$$Y = \sum_{X \in Pa(Y)} a_{xy} X + \xi_Y$$

where all ξ_Y are jointly independent and non-Gaussian distributed. Additionally, we require that for all $Y \in V$, $X \in Pa(Y)$, $a_{xy} \neq 0$. Then, the graph G is identifiable from $P(V)$.

Noise based algorithm

LINGAM

Algorithm 1 LINGAM

Input: $P(\mathcal{V})$

Output: $\mathcal{G}$

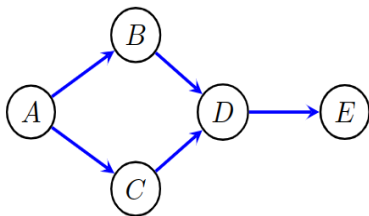
```

1: Form an empty graph  $\mathcal{G}$  on vertex set  $\mathcal{V} = \{X_1, \dots, X_p\}$ 
2: Let  $S = \{1, \dots, p\}$  and  $\mathcal{T} = []$ 
3: repeat
4:    $H = []$ 
5:   for  $i \in S$  do
6:     for  $j \in S \setminus \{i\}$  do
7:        $\hat{\xi}_{ij} = X_j - \frac{\text{cov}(X_i, X_j)}{\text{var}(X_i)} X_i$ 
8:     end for
9:      $h = \sum_{j \in S \setminus \{i\}} \lambda(X_i, \hat{\xi}_{ij})$ 
10:     $H = [H, h]$ 
11:   end for
12:    $i^* = \arg \min_{i \in S} H$ 
13:    $S = S \setminus \{i^*\}$ 
14:    $\mathcal{T} = [\mathcal{T}, i^*]$ 
15:    $\forall j \in S, X_j = \hat{\xi}_{i^*j}$ 
16: until  $|S| = 0$ 
17: Append( $\mathcal{T}, S_0$ )
18: Construct a strictly lower triangular matrix by following the order in  $\mathcal{T}$ , and estimate the connection strengths  $a_{i,j}$  by using some conventional covariance-based regression.
19: if  $a_{i,j} > 0$  then
20:   Add  $X_i \rightarrow X_j$  to  $\mathcal{G}$ 
21: end if
22: Return  $\mathcal{G}$ 

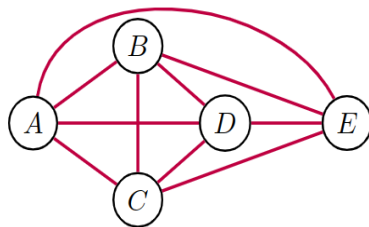
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The PC algorithm

An example



Unknown true graph



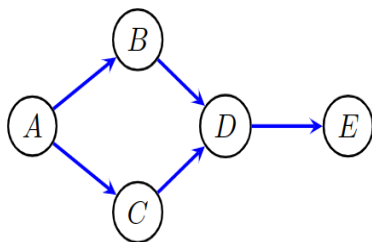
Initial Graph

Figure: The initial graph is complete

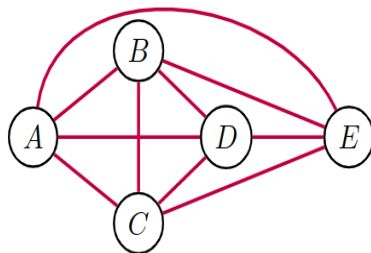
The PC algorithm

An example

Example of PC Algorithm : $k=0$



Unknown true graph



Graph after $k = 0$

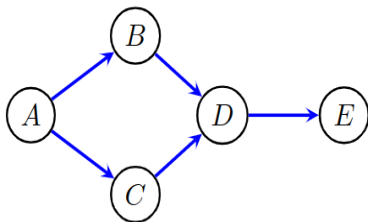
Figure: The initial graph is complete

There is no pair of variables d-separated given $\emptyset$, so the graph is unchanged

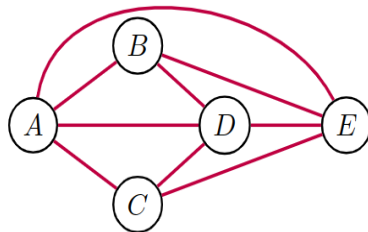
The PC algorithm

An example

Example of PC Algorithm $k = 1$ with (B, C)



Unknown true graph



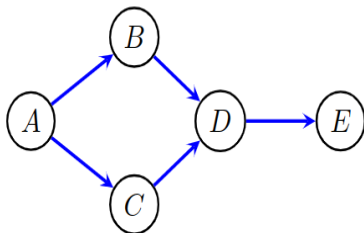
Graph

Since B and C are d-separated given $\{A\}$ we remove the $B - C$ edge and record $S_{BC} = \{A\}$.

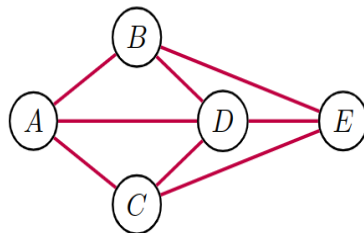
The PC algorithm

An example

Example of PC Algorithm : $k = 1$ with (A, E)



Unknown true graph



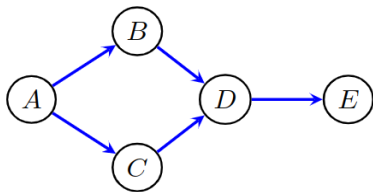
Graph

Since A and E are d separated given $\{D\}$ we remove the $A - E$ edge and record $S_{AE} = \{D\}$.

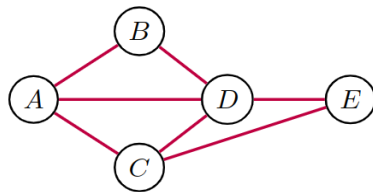
The PC algorithm

An example

Example of PC Algorithm : $k = 1$ with (B, E)



Unknown true graph



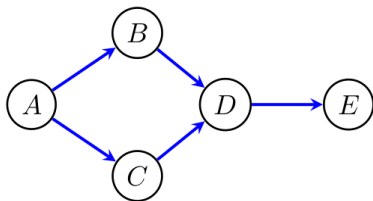
Graph

Since B and E are d separated given $\{D\}$ we remove the $B - E$ edge and record $S_{BE} = \{D\}$.

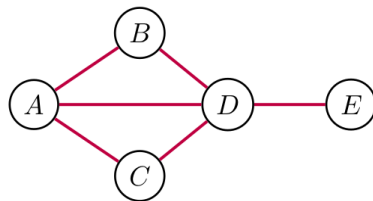
The PC algorithm

An example

Example of PC Algorithm : $k = 1$ with (C, E)



Unknown true graph



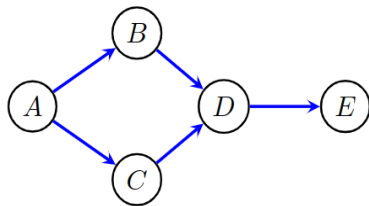
Graph after $k = 1$

Since C and E are d separated given $\{D\}$ we remove the $C - E$ edge and record $S_{CE} = \{D\}$. This completes this stage

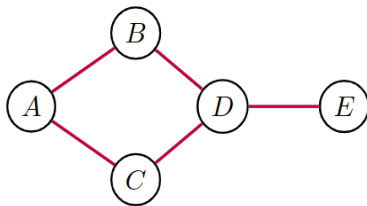
The PC algorithm

An example

Example of PC Algorithm : $k = 2$ with (A, D)



Unknown true graph



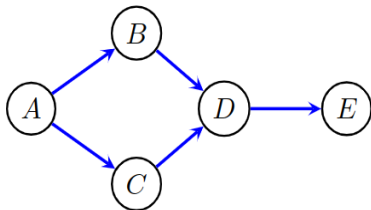
Graph after $k = 2$

Since A and D are d separated given $\{B, C\}$ we remove the $A - D$ edge and record $S_{AD} = \{B, C\}$. This completes this stage.

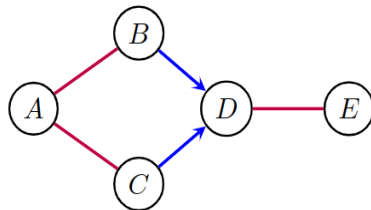
The PC algorithm

An example

Orienting unshielded colliders from separating sets



Unknown true graph



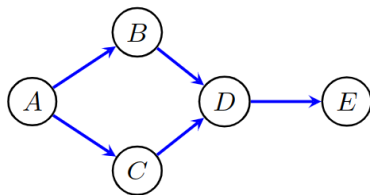
Graph after $k = 1$

Since $D \notin S_{BC} = \{A\}$, we orient $B \rightarrow D \leftarrow C$. The other triples (B, A, C) , (A, B, D) , (A, C, D) , (B, D, E) and (C, D, E) do not lead to further orientation; since the middle vertex is in each separating set.

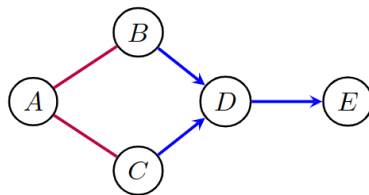
The PC algorithm

An example

Additional orientations to form CPDAG



Unknown true graph



CPDAG

Since (B, D, E) is not a collider, but $B \rightarrow D$ we can orient $D - E$ as $D \rightarrow E$

An example

```

1: INPUT: Vertex Set  $V$ , Conditional Independence Information
2: OUTPUT: Estimated skeleton  $C$ , separation sets  $S$  (only needed when directing the skeleton afterwards)
3: Form the complete undirected graph  $\tilde{C}$  on the vertex set  $V$ .
4:  $\ell = -1$ ;  $C = \tilde{C}$ 
5: repeat
6:    $\ell = \ell + 1$ 
7: repeat
8:   Select a (new) ordered pair of nodes  $i, j$  that are adjacent in  $C$  such that  $|adj(C, i) \setminus \{j\}| \geq \ell$ 
9: repeat
10:  Choose (new)  $\mathbf{k} \subseteq adj(C, i) \setminus \{j\}$  with  $|\mathbf{k}| = \ell$ 
11:  if  $i$  and  $j$  are conditionally independent given  $\mathbf{k}$  then
12:    Delete edge  $i, j$ 
13:    Denote this new graph by  $C$ 
14:    Save  $\mathbf{k}$  in  $S(i, j)$  and  $S(j, i)$ 
15:  end if
16:  until edge  $i, j$  is deleted or all  $\mathbf{k} \subseteq adj(C, i) \setminus \{j\}$  with  $|\mathbf{k}| = \ell$  have been chosen
17:  until all ordered pairs of adjacent variables  $i$  and  $j$  such that  $|adj(C, i) \setminus \{j\}| \geq \ell$  and  $\mathbf{k} \subseteq adj(C, i) \setminus \{j\}$  with  $|\mathbf{k}| = \ell$  have been tested for conditional independence
18: until for each ordered pair of adjacent nodes  $i, j$ :  $|adj(C, i) \setminus \{j\}| < \ell$ .

```

The PC algorithm

An example

Algorithm 2 Extending the skeleton to a CPDAG

INPUT: Skeleton G_{skel} , separation sets S

OUTPUT: CPDAG G

for all pairs of nonadjacent variables i, j with common neighbour k **do**

if $k \notin S(i, j)$ **then**

 Replace $i - k - j$ in G_{skel} by $i \rightarrow k \leftarrow j$

end if

end for

In the resulting PDAG, try to orient as many undirected edges as possible by repeated application of the following three rules:

R1 Orient $j - k$ into $j \rightarrow k$ whenever there is an arrow $i \rightarrow j$ such that i and k are nonadjacent.

R2 Orient $i - j$ into $i \rightarrow j$ whenever there is a chain $i \rightarrow k \rightarrow j$.

R3 Orient $i - j$ into $i \rightarrow j$ whenever there are two chains $i - k \rightarrow j$ and $i - l \rightarrow j$ such that k and l are nonadjacent.

R4 Orient $i - j$ into $i \rightarrow j$ whenever there are two chains $i - k \rightarrow l$ and $k \rightarrow l \rightarrow j$ such that k and l are nonadjacent.

Noise based algorithm

Theorem (ANM)

Assume a linear SCM with graph $G = (V, E)$ and a compatible distribution $P(V)$ such that for all $Y \in V$

$$Y = f((X \in Pa(Y)) + \xi_Y$$

where all ξ_Y are jointly independent. Then, the graph G is identifiable from $P(V)$.

- └ Additionnal stuff
- └ A non linear extension of LINGAM

Noise based algorithm

ANM

Algorithm 2 ANM

Input: $P(\mathcal{V})$

Output: $\mathcal{G}$

```

1: Form an empty graph  $\mathcal{G}$  on vertex set  $\mathcal{V} = \{X_1, \dots, X_p\}$ 
2: Let  $S = \{1, \dots, p\}$  and  $\mathcal{T} = []$ 
3: repeat
4:    $H = []$ 
5:   for  $j \in S$  do
6:      $\hat{f}_j$ : Regress  $X_j$  on  $\{X_i\}_{i \in S \setminus \{j\}}$ 
7:      $\hat{\epsilon}_j = X_j - \hat{f}_j(X_i)$ 
8:      $h = \hat{\gamma}(\{X_i\}_{i \in S \setminus \{j\}}, \hat{\epsilon}_j)$ 
9:      $H = [H, h]$ 
10:  end for
11:   $i^* = \arg \min_{i \in S} H$ 
12:   $S = S \setminus \{i^*\}$ 
13:   $\mathcal{T} = [i^*, \mathcal{T}]$ 
14: until  $|S| = 0$ 
15: for  $j \in \{2, \dots, p\}$  do
16:   for  $i \in \{\mathcal{T}_1, \dots, \mathcal{T}_{j-1}\}$  do
17:      $\hat{f}_i$ : Regress  $X_i$  on  $\{X_k\}_{k \in \{\mathcal{T}_1, \dots, \mathcal{T}_{j-1}\} \setminus \{i\}}$ 
18:      $\hat{\epsilon}_i = X_i - \hat{f}_i(X_k)$ 
19:     if  $\{X_k\}_{k \in \{\mathcal{T}_1, \dots, \mathcal{T}_{j-1}\} \setminus \{i\}} \not\perp_P \hat{\epsilon}_i$  then
20:       Add  $X_i \rightarrow X_j$  to  $\mathcal{G}$ 
21:     end if
22:   end for
23: end for
24: Return  $\mathcal{G}$ 

```
